17º MeetUp de PyData
Granada, 14/05/2026

Flujo de trabajo en Cienciometría

Del dato al resultado de investigación

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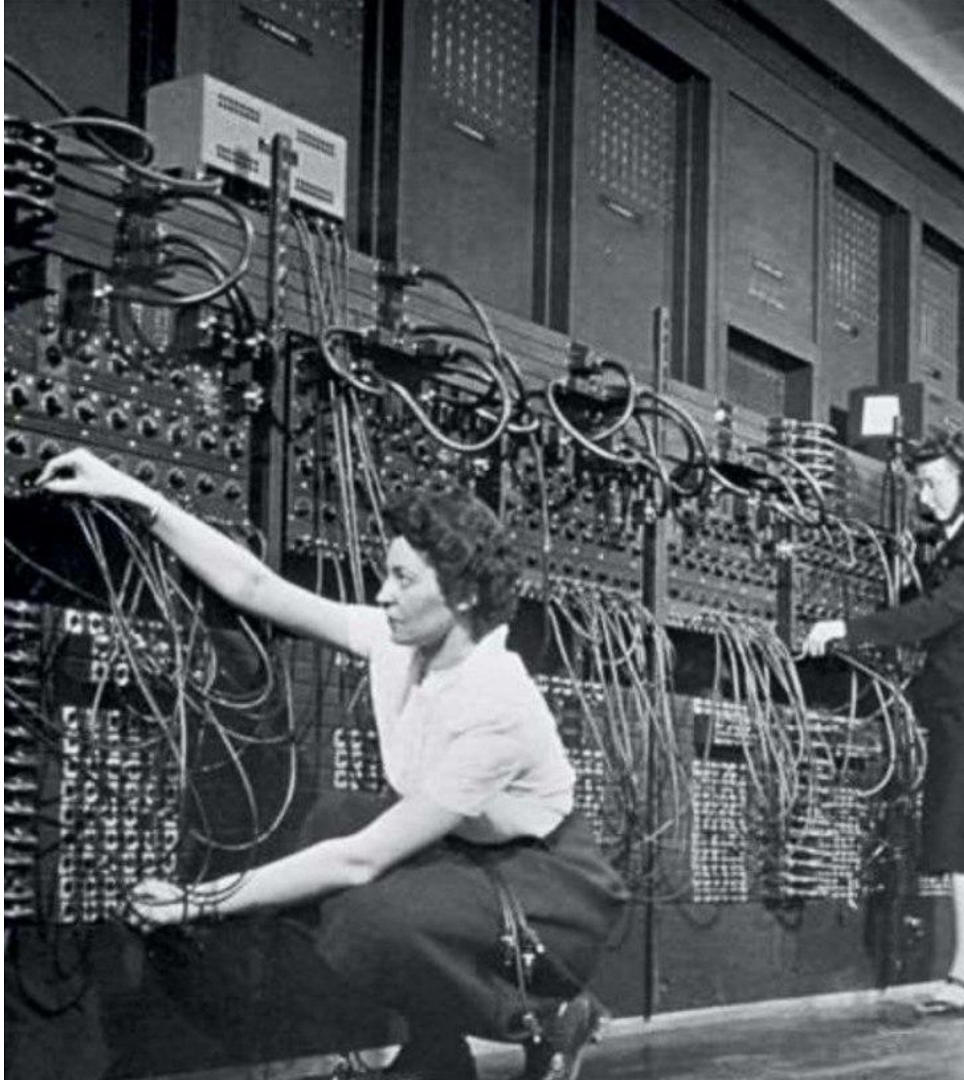
Ciencia de la Ciencia

- Sociología de la Ciencia
- Historia de la Ciencia
- Filosofía de la Ciencia
- Política Científica
- Estudios de la Tecnología y de la Innovación
- Economía de la I+D
- Comunicación Científica
- …
- **Cienciometría**

Flujo de trabajo en Cienciometría

1. Obtención de datos
2. Procesamiento o limpieza
3. Combinación de fuentes
4. Cálculo de variables e indicadores
5. Generación de resultados en tablas y figuras
6. …
7. Difusión de datasets y código



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Artículo

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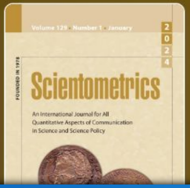

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Elvira González-Salmón, Victoria Di César, Aoxia Xiao & Nicolas

Preprint

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Published February 17, 2026 | Version v1 Preprint Open

Presence and visibility of local journals in international scientific databases

Di César, Victoria [1]; Robinson-Garcia, Nicolas [1] Show affiliations

Mainstream databases present significant language, geographical, disciplinary, and thematic biases, particularly against research addressing local needs. Evaluation policies framed around mainstream-based indicators tend to reproduce these biases and, consequently, underestimate local research. We address this issue empirically by merging four databases and conducting a large-scale analysis of local journals across mainstream (WOS and Scopus) and non-mainstream (OpenAlex and DOAJ) knowledge circuits. We aim to (1) determine the representation of local journals in the four databases, (2) assess the visibility of the fields and countries publishing in them, and (3) examine their non-English and open access features. We build a dataset of more than 75,000 OpenAlex journals, which are enriched with WOS, Scopus, and DOAJ variables and characterized as local or global according to a local research framework. Results show that most local journals are poorly represented in the mainstream, regardless of their field, language, or access type. Countries publishing in non-English and open access local journals do not follow clear patterns when examined from different locality perspectives. One conclusion is straightforward: local journals circulate primarily outside the mainstream. From a policy perspective, local research should not be discussed and evaluated solely from the content covered by WOS and Scopus.

Notes (English)

Versions

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10.5281/zenodo.18671033

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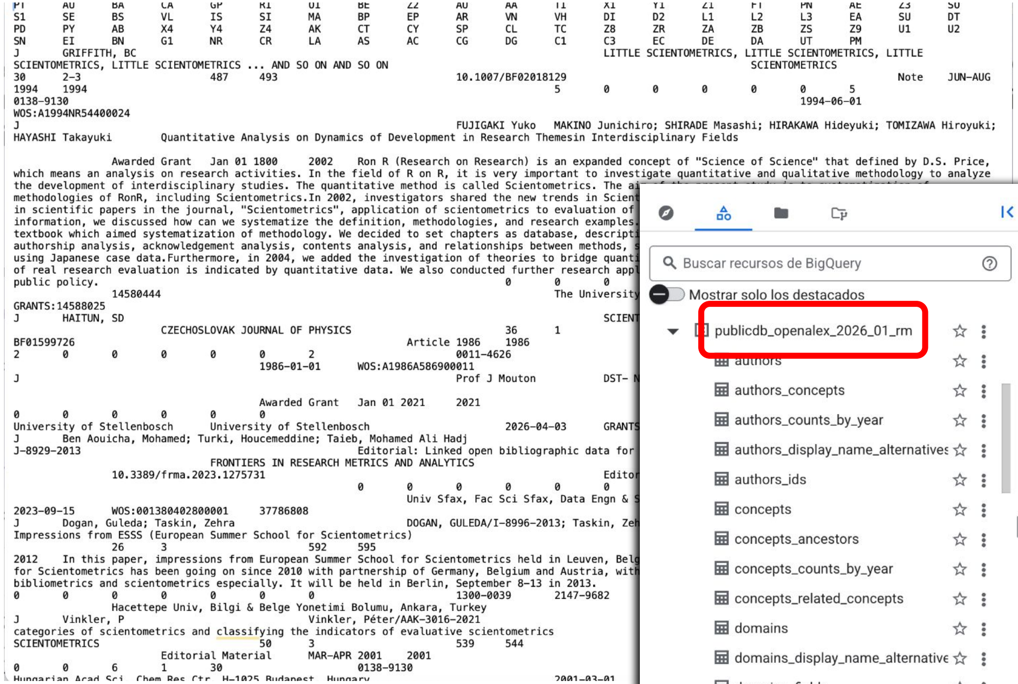
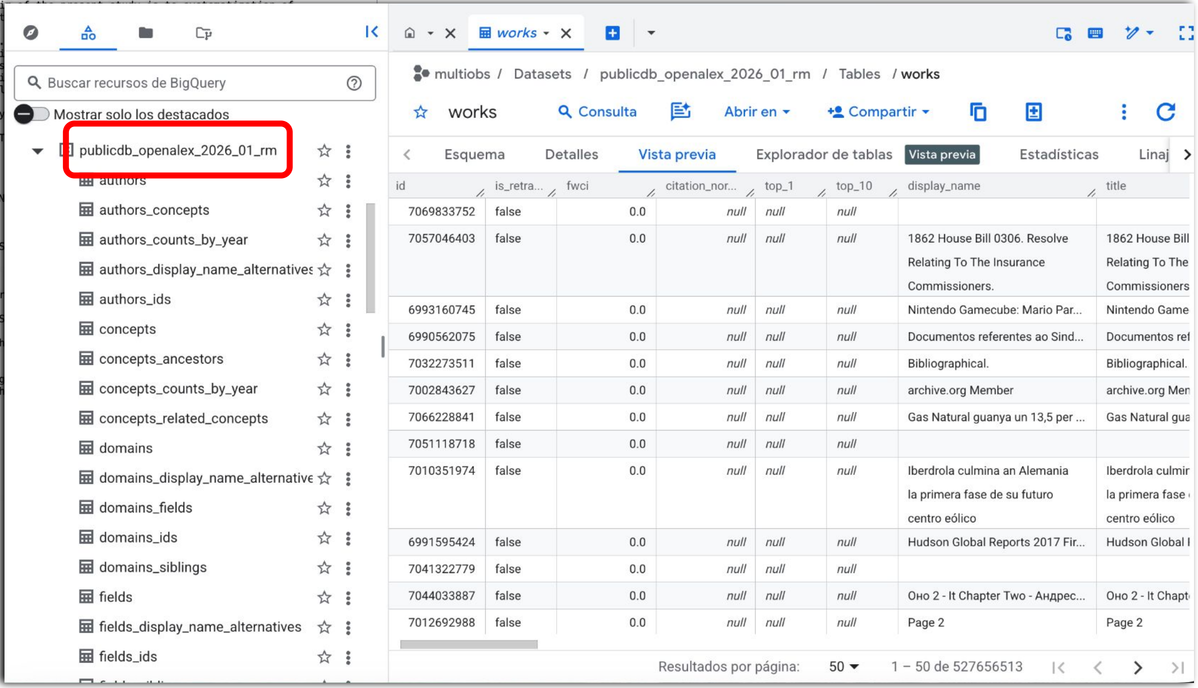


OpenAlex

DOAJ



Clarivate Web of Science™

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work_id	author_id	raw_affiliation_string
2086444734	5073170033	CIS Project Officer, European Defence Agency, France
1600193241	5024361021	Department of Biology, University of California, San Diego., USA
2089113425	5052703097	Department of Otorhinolaryngology: Head and Neck Surgery, University of Pennsylvania, Philadelphia, Pennsylvania 19104 USA
4367331826	5024480921	Department of Surgery, National University Hospital of Singapore, Singapore
2588920864	5065769358	Uniwersytet Ekonomiczny w Krakowie,\nKatedra Metrologii i Analizy Instrumentalnej
2036821989	5014032103	Institut de Biotechnologie, Université de Limoges, 123, avenue Albert Thomas, 87060 Limoges Cedex, France
1969245491	5003925486	Wellcome Research Laboratories, Beckenham, Kent
4398977521	5034916051	(University of Tehran)
4398983648	5006144753	(Princeton University and International Crisis Grou
4239070112	5063370691	Smt. G. R. Doshi and Smt. K. M. Mehta Institute of Ahmedabad, India
2024077717	5007586601	Inst. de Physique, Neuchatel Univ., Switzerland
4213192936	5022205554	NTU, Vietnam
60880552	5103401424	Texas A&M University, Institute of Biosciences and
4213144618	5027179585	University
3217587195	5024309554	Flemming J Olsen Gentofte Hosp, Valby
4237969420	5071290274	TH Nürnberg Georg Simon Ohm, Nürnberg, Deuts
1990578475	5113841906	2Syracuse University, Syracuse, NY.
4313428599	5009196289	Universidad Autónoma de Baja California
4250276411	5113986281	Stanford University
888760880	5011648613	Los Angeles CA United States
4400968851	5006191066	Environmental Molecular Sciences Laboratory
4392263796	5034798259	CREGO - Centre de Recherche en Gestion des Org

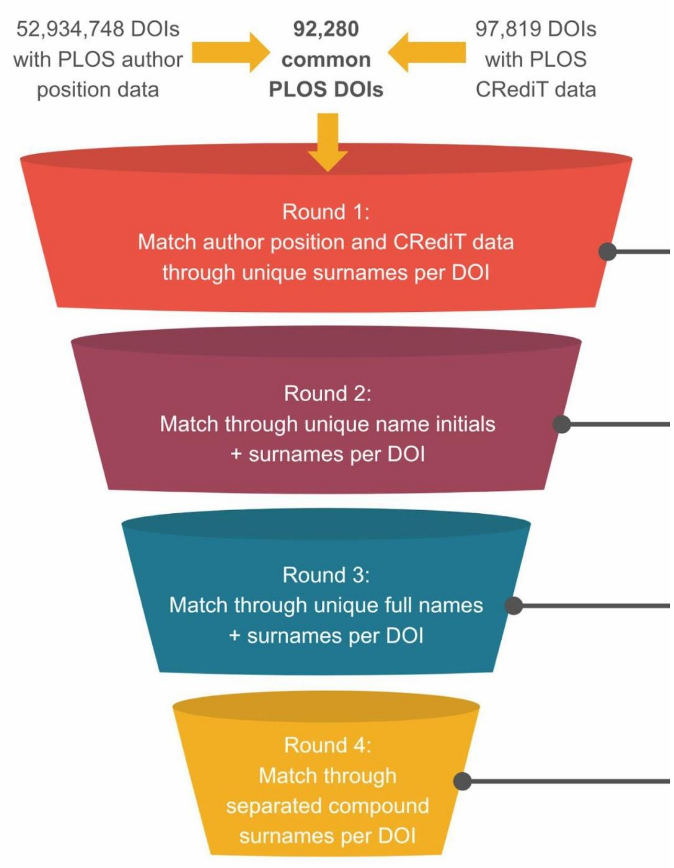
		19104
3015237345	5017623751	University of Pennsylvania, Philadelphia, Pennsylvania
2024139507	5076466572	University Children's Hospital "Het Wilhelmina Kinderziekenhuis" and the

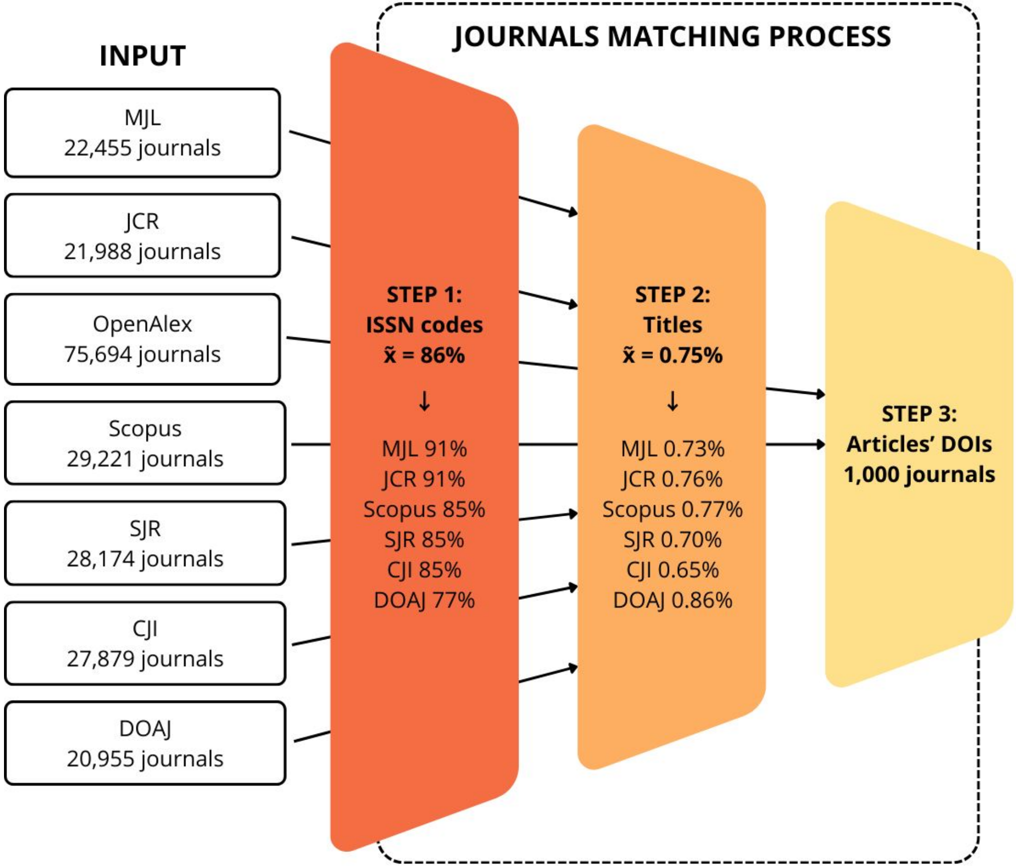
...3662122	5015070406	Department of Physics, University of California, Santa Barbara, California 93106
...4973047	5051869254	Department of Chemistry and Biochemistry and Center for Nanoscience, University of Missouri—St. Louis, One University Boulevard, St. Louis, Missouri 63121, United States

		One UTSA Circle, San Antonio, TX, 78249, USA
2121344482	5044544552	Dept. of Electrical and Computer Engineering, University of California, San Diego, CA 92093, USA

Google Big Query

Google Cloud Storage


52,934,748 DOIs with PLOS author position data
92,280 common PLOS DOIs
97,819 DOIs with PLOS CRediT data
Round 1: Match author position and CRediT data through unique surnames per DOI
Round 2: Match through unique name initials + surnames per DOI
Round 3: Match through unique full names + surnames per DOI
Round 4: Match through separated compound surnames per DOI

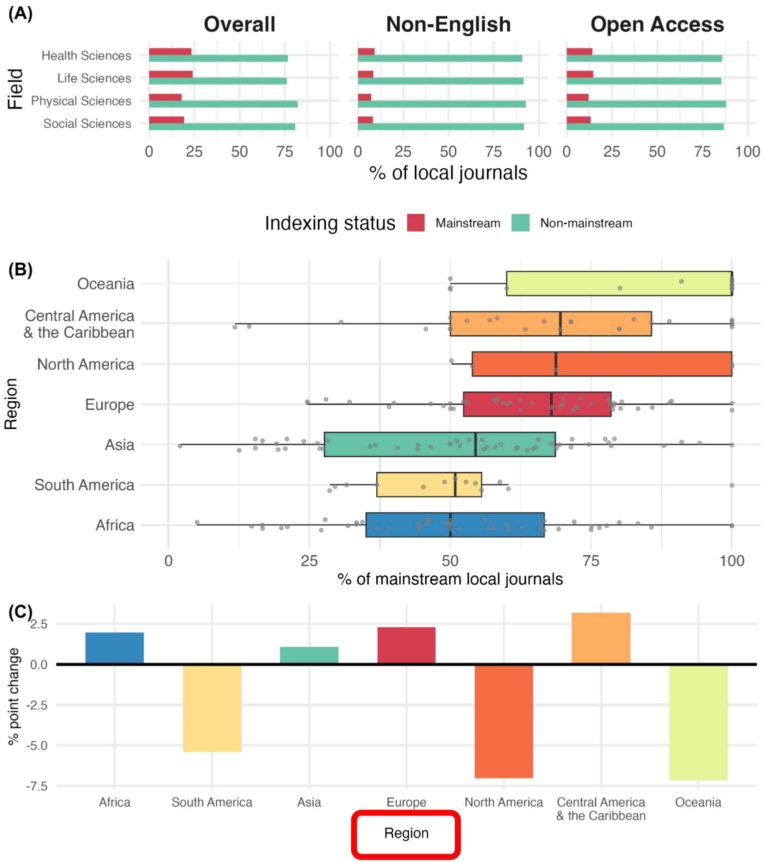

INPUT
MJL 22,455 journals
JCR 21,988 journals
OpenAlex 75,694 journals
Scopus 29,221 journals
SJR 28,174 journals
CJI 27,879 journals
DOAJ 20,955 journals
JOURNALS MATCHING PROCESS
STEP 1: ISSN codes x̃ = 86%
MJL 91%
JCR 91%
Scopus 85%
SJR 85%
CJI 85%
DOAJ 77%
STEP 2: Titles x̃ = 0.75%
MJL 0.73%
JCR 0.76%
Scopus 0.77%
SJR 0.70%
CJI 0.65%
DOAJ 0.86%
STEP 3: Articles' DOIs 1,000 journals

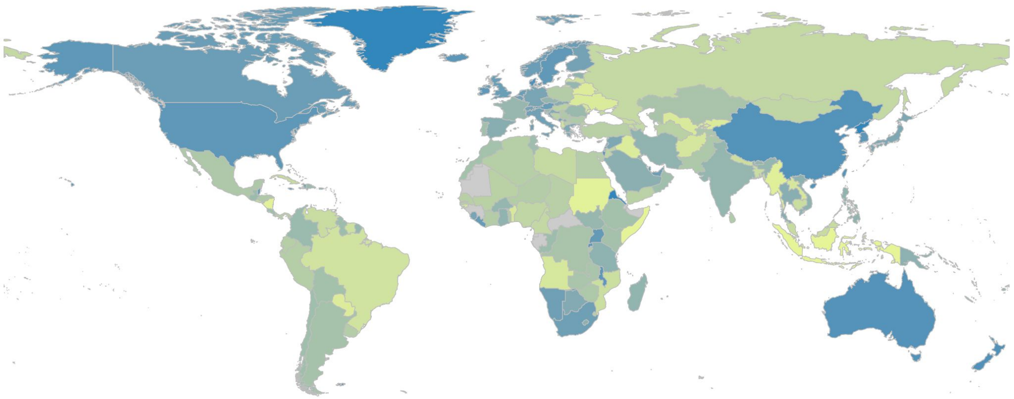
```
### LOCAL VARIABLES COMPUTATION
## REFERENCES
# read files and split into 20 dataframes for processing
references_files <- list.files(path = "~/Desktop/OpenAlex_journals_dataset/references_local_variable", pattern = "^local_research_OA2410_references_1
num_parts <- 20
num_files <- length(references_files)
chunk_size <- ceiling(num_files / num_parts)
for (i in 1:num_parts) {chunk_files <- references_files[((i - 1)
                        chunk_files <- chunk_files[!is.na(chunk_f
                        chunk_data <- rbindlist(lapply(chunk_file
                        assign(paste0("references_part_", i), chu
                        rm(chunk_data)
                        gc()}

# compute the refs_count and refs_total variables recurrently pe
references_part_1 <- references_part_1 %>% group_by(journal_id, j
                                mutate(refs_count = n(
                                ungroup()
references_part_1 <- within(references_part_1, rm(article_id, ref
references_part_1 <- references_part_1 %>% distinct()
references_part_1 <- references_part_1 %>% group_by(journal_id, j
                                mutate(refs_total = su
                                ungroup()

# merge all parts together without loosing rows
local_variable_refs <- rbind(references_part_1, references_part_2
                          references_part_7, references_part_8
                          references_part_13, references_part_
                          references_part_19, references_part_

# compute variables refs_count, refs_total and refs_prop per uni
local_variable_refs <- local_variable_refs %>% group_by(journal_i
                                    summarise
local_variable_refs <- local_variable_refs %>% filter(!(journal_i
local_variable_refs <- local_variable_refs %>% group_by(journal_i
                                    mutate(re
                                    ungroup(
local_variable_refs <- local_variable_refs %>% group_by(journal_i
                                    filter(re
                                    ungroup(
```

Filter | Cols: « < 55 - 104 > »

refs_prop	refs_country	cits_prop	cits_country	pubs_prop	pubs_country	tops_prop	langs	mains	local_territory	local_producers	local_recipients
0.23	United States	0.16	United States	0.29	United States	0.00	1	1	global	global	global
0.22	United States	0.10	United States	0.11	Iran	0.18	1	1	local	global	global
0.26	United States	0.14	China	0.23	China	0.08	1	1	global	global	global
0.39	United States	0.34	United States	0.61	United States	0.07	1	1	global	global	global
0.26	China	0.20	China	0.38	China	0.00	1	1	global	global	global
0.29	Brazil	0.60	Brazil	0.75	Brazil	0.05	0	1	global	global	global
0.18	United States	0.16	China	0.50	Austria	0.06	1	1	global	global	global
0.39	United States	0.41	United States	0.41	United States	0.29	1	1	local	global	global
0.44	United States	0.32	United States	0.33	United States	0.00	1	1	global	local	global
0.39	United States	0.57	China	0.43	China	0.00	1	1	global	global	global
0.78	United States	0.29	United States	1.00	United States	0.00	1	1	global	local	global
0.26	United States	0.25	Japan	0.68	Japan	0.06	1	1	global	global	global
0.36	United States	0.18	United States	0.21	United States	0.00	1	1	global	global	global
0.32	United States	0.19	United States	0.28	United States	0.00	1	1	global	global	global
0.51	Russia	0.73	Russia	0.90	Russia	0.01	1	1	global	local	global
0.18	India	0.43	India	0.57	India	0.06	1	1	global	global	global
0.25	United States	0.15	Brazil	0.22	Brazil	0.21	1	1	local	global	global
0.22	United States	0.15	United States	0.36	Turkey	0.04	1	1	global	global	global
0.24	United States	0.20	Japan	0.98	Japan	0.08	1	1	global	global	global
0.21	China	0.24	China	0.37	China	0.00	1	1	global	global	global
0.29	United States	0.19	China	0.24	United States	0.02	1	1	global	global	global
0.26	United States	0.33	South Korea	0.95	South Korea	0.01	1	1	global	global	global
0.31	United States	0.24	China	0.23	United States	0.14	1	1	local	global	global
0.24	United States	0.18	United States	0.27	United States	0.01	1	1	global	global	global
0.15	China	0.16	China	0.16	China	0.01	1	1	global	global	global
0.22	United States	0.20	United States	0.30	United Kingdom	0.28	1	1	local	global	global

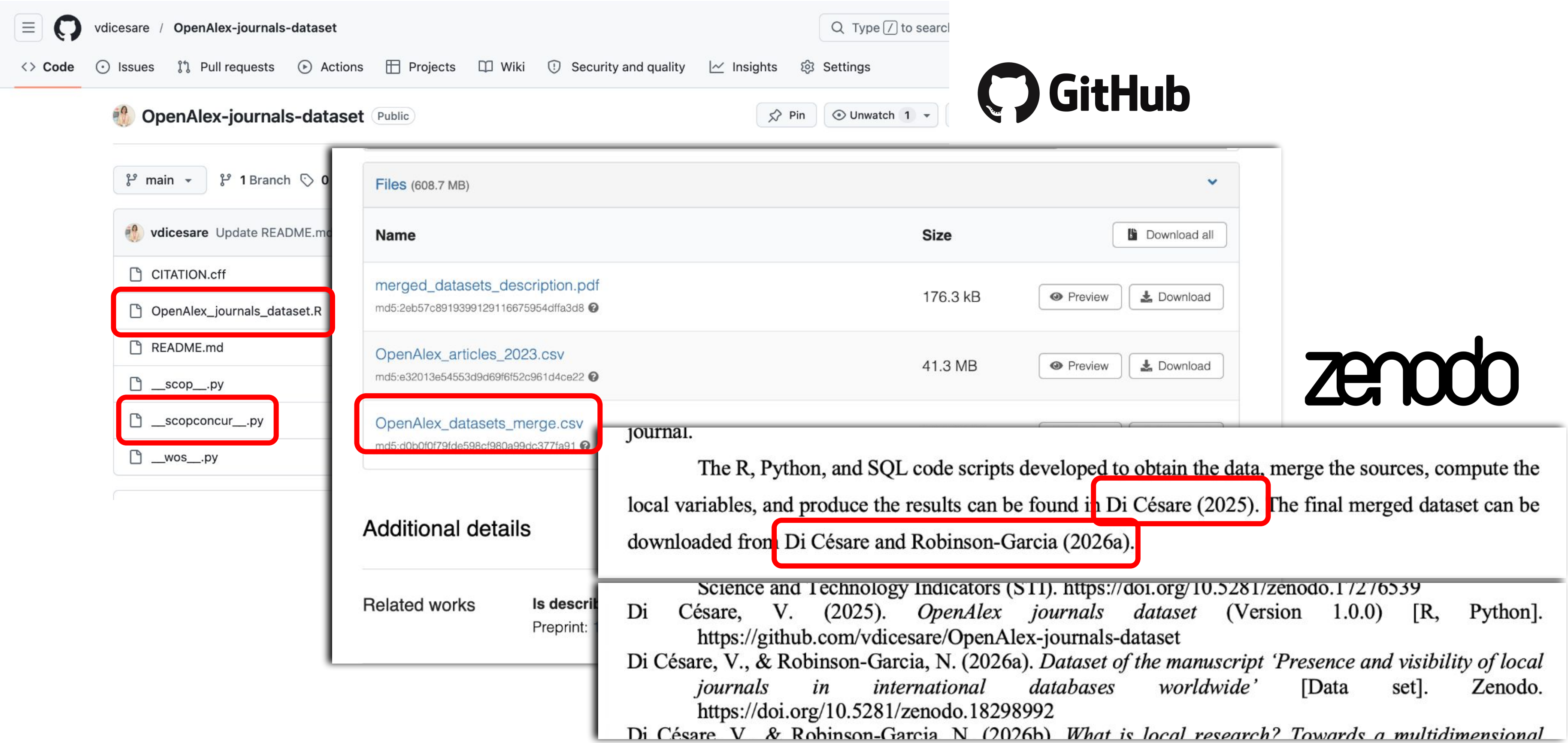
owing 1 to 25 of 109,652 entries, 104 total columns





Database	Field	Number of journals					
		Total	Locally situated	Locally informed	Locally relevant	Non-English	Open Access
OpenAlex	**Overall**	**75,694**	**19,766**	**16,872**	**15,004**	**11,236**	**30,012**
	Health Sciences	16,278	3,987	3,107	2,924	2,231	6,537
	Life Sciences	13,804	3,583	2,334	2,348	1,710	5,724
	Physical Sciences	21,357	5,459	4,801	4,370	3,045	8,462
	Social Sciences	24,255	6,737	6,630	5,362	4,250	9,289

Obtención | Procesamiento | Combinación | Cálculo | Resultados | **Difusión**



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¡Muchas gracias!

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